Quiz#4

Name(print): _____ **Student ID:** _____

1. A Chinook salmon can jump out of water with a speed v_0 and an angle θ. What is the speed of the fish when it is at the maximum height?
   A. v_0.
   B. $v_0 \cos\theta$.
   C. $v_0 \sin\theta$.
   D. 0.

2. A spaceship is traveling with an initial velocity of $\vec{v}_0 = (20.1\ m/s)\hat{\imath}$ when its rockets fire, imparting an acceleration of $\vec{a} = (2.02\,m/s^2)\hat{\imath} + (4.03\,m/s^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?
   A. The x-component of the spaceship's velocity remains constant.
   B. The y-component of the spaceship's velocity remains constant.
   C. Both the direction of motion and speed change.
   D. None of the above is correct.

Using the five-step problem-solving strategy, solve the following problem:

3. A stone is launched into the air from the top of a building with an initial speed of 6.7 m/s at an angle of 60° above the horizontal. How much time does it take for the stone to reach its maximum height?